Network Project Report

Student Information

Name	Section	Bench Number
Student 1	Sec	BN1
Student 2	Sec	BN2

Date: June 3, 2025

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1 Effect of TCP Window Size

1.1 Question 2.i

The Throughout Appears to be increasing with Window Size (as Expected) with Zero Retransmissions up to a certain limit where the receiver Buffer gets Full at 6KB and tries to increase it's buffer size so performance is increases again thus making a zigzag type of graph, then at 24KB the Buffer can’t get any larger so Performance is Poor.

$$Throughout = \frac{Data\ Transferred}{Interval\ Time}$$

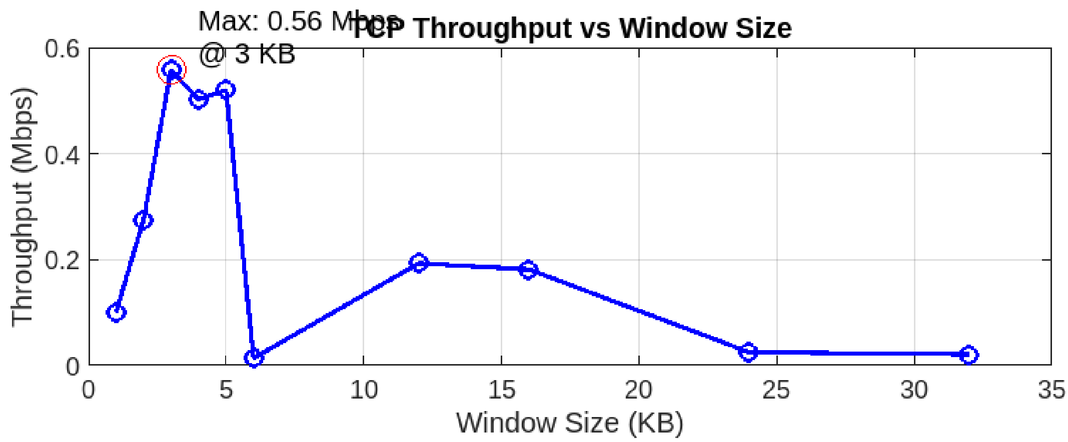


Figure 1: TCP Window Size vs Throughput

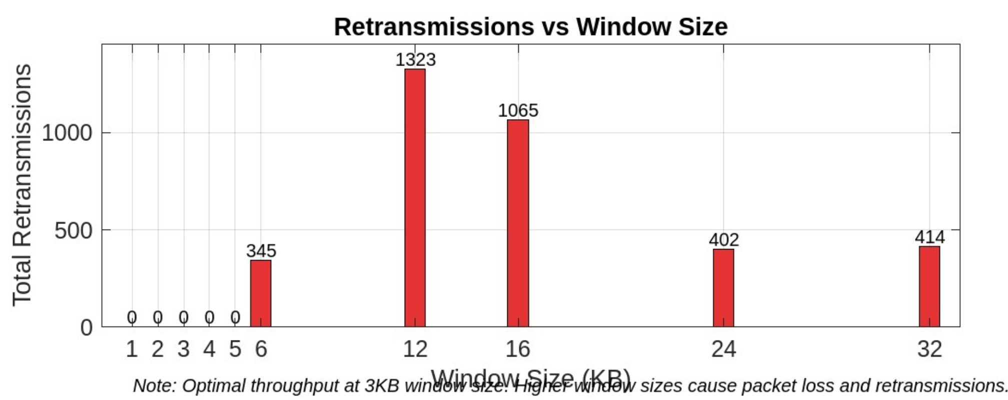


Figure 2: TCP Window Size vs Retransmissions

Window Size (KB)	Data (MB)	Throughput (MB/s)	Retransmissions
1	4.01	0.10025	0
2	11.00	0.27500	0
3	22.30	0.55750	0
4	20.10	0.50250	0
5	20.80	0.52000	0
6	0.58	0.01450	345
12	7.73	0.19325	1323
16	7.27	0.18175	1065
24	1.022	0.02555	402
32	0.853	0.021325	404

Table 1: TCP Window Size vs Throughput Data

1.2 Question 2.ii

Frame length is 76 bytes
IPv4 Header length : 20 bytes
TCP header length : 40 bytes
Ethernet = 16 bytes

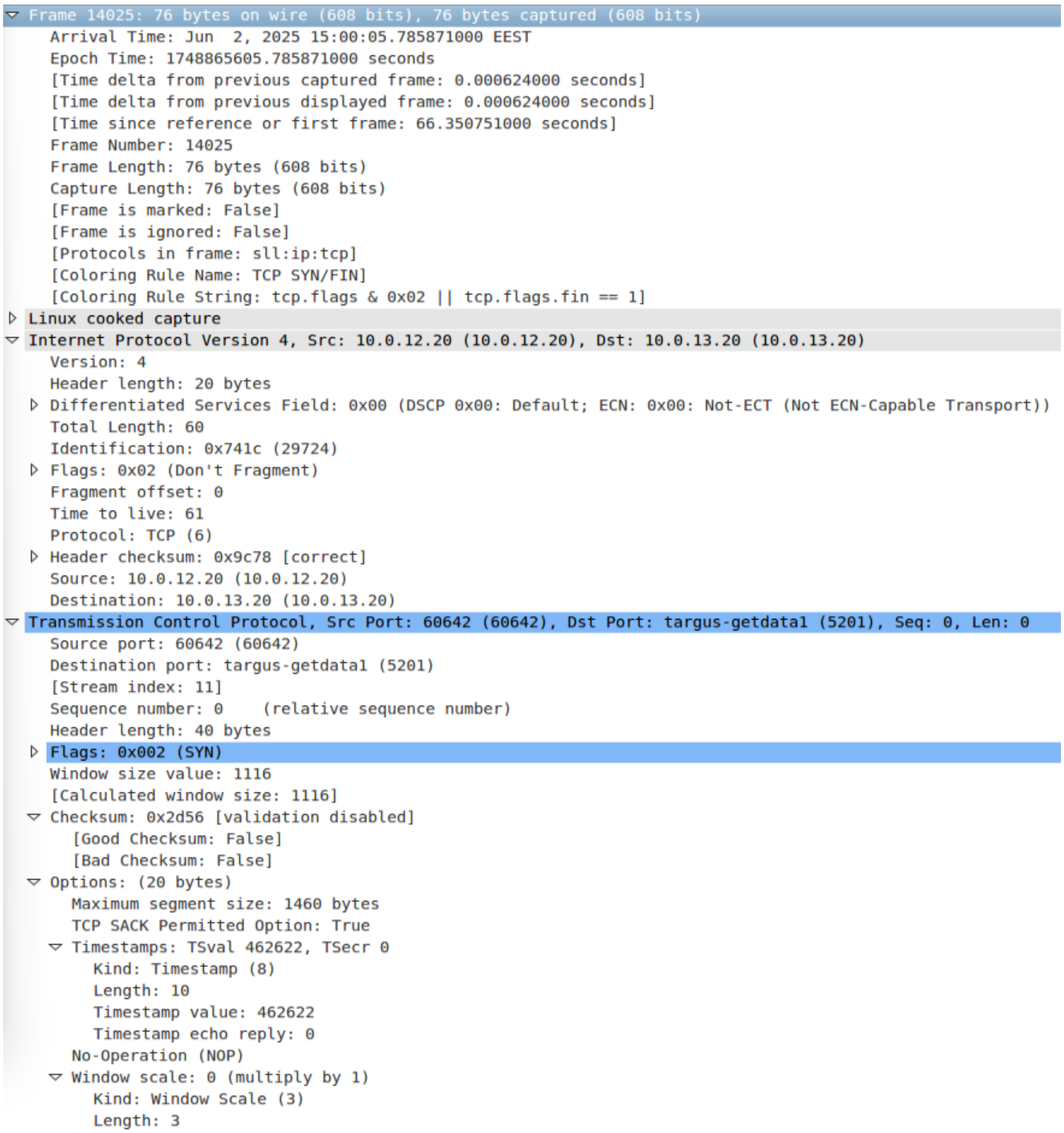


Figure 3: Sent packet from node n7

1.3 Question 2.iii

Frame length is 68 bytes
IPv4 Header length : 20 bytes
TCP header length : 32 bytes
Ethernet = 16 bytes
Ack number = 42 & Sequance No. = 2 (Both Relative)

```
▽ Frame 13993: 68 bytes on wire (544 bits), 68 bytes captured (544 bits)
    Arrival Time: Jun  2, 2025 15:00:05.778537000 EEST
    Epoch Time: 1748865605.778537000 seconds
    [Time delta from previous captured frame: 0.000669000 seconds]
    [Time delta from previous displayed frame: 0.000669000 seconds]
    [Time since reference or first frame: 66.343417000 seconds]
    Frame Number: 13993
    Frame Length: 68 bytes (544 bits)
    Capture Length: 68 bytes (544 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: sll:ip:tcp]
    [Coloring Rule Name: Bad TCP]
    [Coloring Rule String: tcp.analysis.flags]
▷ Linux cooked capture
▽ Internet Protocol Version 4, Src: 10.0.13.20 (10.0.13.20), Dst: 10.0.12.20 (10.0.12.20)
    Version: 4
    Header length: 20 bytes
  ▷ Differentiated Services Field: 0x00 (DSCP 0x00: Default; ECN: 0x00: Not-ECT (Not ECN-Capable Transport))
    Total Length: 52
    Identification: 0xfecc (65228)
  ▷ Flags: 0x02 (Don't Fragment)
    Fragment offset: 0
    Time to live: 61
    Protocol: TCP (6)
  ▷ Header checksum: 0x11d0 [correct]
    Source: 10.0.13.20 (10.0.13.20)
    Destination: 10.0.12.20 (10.0.12.20)
▽ Transmission Control Protocol, Src Port: targus-getdata1 (5201), Dst Port: 60641 (60641), Seq: 2, Ack: 42, Len: 0
    Source port: targus-getdata1 (5201)
    Destination port: 60641 (60641)
    [Stream index: 10]
    Sequence number: 2    (relative sequence number)
    Acknowledgement number: 42    (relative ack number)
    Header length: 32 bytes
  ▷ Flags: 0x010 (ACK)
    Window size value: 905
    [Calculated window size: 14480]
    [Window size scaling factor: 16]
  ▽ Checksum: 0x2d4e [validation disabled]
      [Good Checksum: False]
      [Bad Checksum: False]
  ▽ Options: (12 bytes)
      No-Operation (NOP)
      No-Operation (NOP)
    ▽ Timestamps: TSval 462621, TSecr 462611
        Kind: Timestamp (8)
        Length: 10
        Timestamp value: 462621
        Timestamp echo reply: 462611
  ▽ [SEQ/ACK analysis]
    ▷ [TCP Analysis Flags]
```

Figure 4: Received from node n11

Window Size (KB)	Data (MB)	Throughput (MB/s)
4(old)	20.10	0.50250
4(new)	7.28	0.182

Table 2: TCP With Different Propagation Delay

2 TCP Short versus Long Paths

2.1 Question 3.i

The Lower Performance is due to larger Delay between n7 & n8 ($4 * 160 + 2 * 110\mu$sec) compared to n7 & n11($3 * 160 + 110\mu$sec)

Terminal Output:

```
64 bytes from 10.0.11.20: icmp_req=3 ttl=59 time=6.04 ms
^Z
[1]+  Stopped                 ping 10.0.11.20
n7.conf# iperf3 -c 10.0.11.20 -t 40 -i 10 -w 4k
Connecting to host 10.0.11.20, port 5201
[4] local 10.0.12.20 port 48635 connected to 10.0.11.20 port 5201
[ID] Interval           Transfer     Bandwidth       Retr  Cwnd
[4]   0.00-10.00  sec  2.00 MBytes  1.68 Mbits/sec    1   8.48 KBytes
[4]  10.00-20.00  sec  1.71 MBytes  1.44 Mbits/sec    0   8.48 KBytes
[4]  20.00-30.12  sec  1.63 MBytes  1.35 Mbits/sec    0   8.48 KBytes
[4]  30.12-40.00  sec  1.94 MBytes  1.64 Mbits/sec    4   8.48 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ID] Interval           Transfer     Bandwidth       Retr
[4]   0.00-40.00  sec  7.28 MBytes  1.53 Mbits/sec    5             sender
[4]   0.00-40.00  sec  7.27 MBytes  1.53 Mbits/sec                  receiver
iperf Done.
```

3 Higher Link Capacity with Drops versus Reliable Lower Capacity

Case #	Data (MB)	Throughput (MB/s)	Throughput (Mbps)	Retransmissions
A	10.10	0.2525	2.13	4
B	9.91	0.2478	2.08	2
C	2.09	0.0523	0.438	108
D	0.728	0.0182	0.149	91
E	6.70	0.1675	1.41	56
F	8.74	0.2185	1.83	25

Table 3: 4KB Window Size Performance Variability (40-second tests)

3.1 Question 4.i

Throughout in case A is the best among these results due to high rate and no loss

Also Case B is better Than C as the Bandwidth isn't really an issue with this window Size So Loss from both directions Really Impacts the Performance of Retransmissions and thus the throughput

3.2 Question 4.ii

Case B is better as there's no loss with adequate Capacity For given window Size

3.3 Question 4.iii

Case E is better Loss in the direction from n4 to n5 is Just Dropped Acks which doesn't effect the throughput that much especially in Go-BACK-N protocol

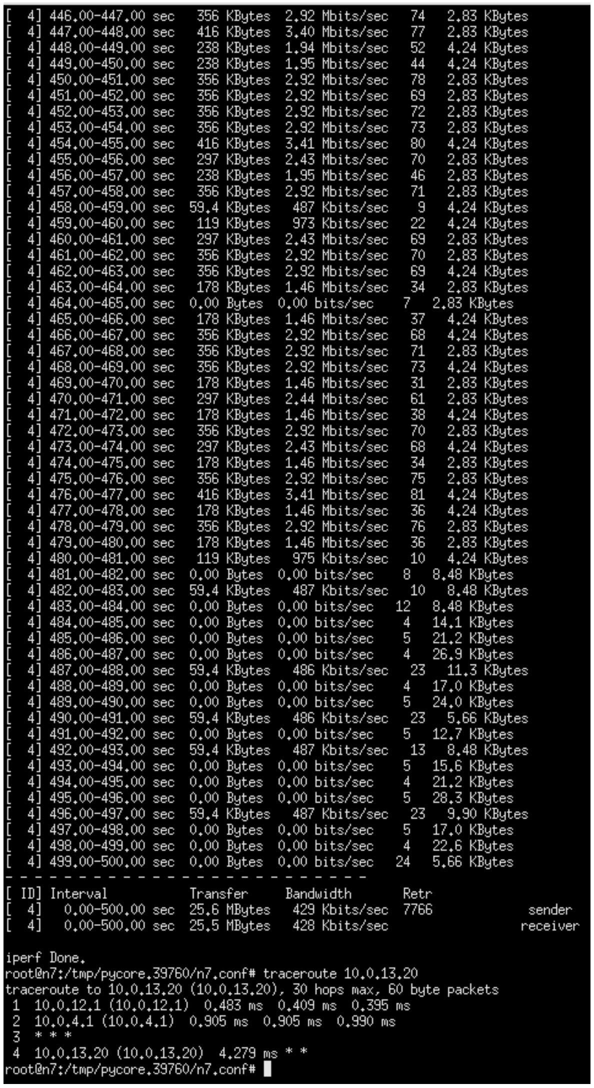
4 OSPF Link Cost Changes

4.1 Question 5.i

```
[  4] 446.00-447.00 sec  19.8 KBytes   162 Kbits/sec    4   22.6 KBytes
[  4] 447.00-448.00 sec  0.00 Bytes  0.00 bits/sec    5   29.7 KBytes
[  4] 448.00-449.00 sec  0.00 Bytes  0.00 bits/sec    4   35.4 KBytes
[  4] 449.00-450.00 sec  59.4 KBytes   487 Kbits/sec   28    9.90 KBytes
[  4] 450.00-451.00 sec  0.00 Bytes  0.00 bits/sec    5   17.0 KBytes
[  4] 451.00-452.00 sec  0.00 Bytes  0.00 bits/sec    4   22.6 KBytes
[  4] 452.00-453.00 sec  0.00 Bytes  0.00 bits/sec    4   28.3 KBytes
[  4] 453.00-454.00 sec  62.2 KBytes   510 Kbits/sec   26    5.66 KBytes
[  4] 454.00-455.00 sec  0.00 Bytes  0.00 bits/sec    4   11.3 KBytes
[  4] 455.00-456.00 sec  0.00 Bytes  0.00 bits/sec    5   18.4 KBytes
[  4] 456.00-457.00 sec  0.00 Bytes  0.00 bits/sec    5   24.0 KBytes
[  4] 457.00-458.00 sec  0.00 Bytes  0.00 bits/sec    4   31.1 KBytes
[  4] 458.00-459.00 sec  59.4 KBytes   486 Kbits/sec   27    7.07 KBytes
[  4] 459.00-460.01 sec  0.00 Bytes  0.00 bits/sec    4   12.7 KBytes
[  4] 460.01-461.00 sec  0.00 Bytes  0.00 bits/sec    5   19.8 KBytes
[  4] 461.00-462.00 sec  0.00 Bytes  0.00 bits/sec    5   25.5 KBytes
[  4] 462.00-463.00 sec  0.00 Bytes  0.00 bits/sec    4   32.5 KBytes
[  4] 463.00-464.00 sec  59.4 KBytes   486 Kbits/sec   26    9.90 KBytes
[  4] 464.00-465.00 sec  0.00 Bytes  0.00 bits/sec    5   17.0 KBytes
[  4] 465.00-466.00 sec  0.00 Bytes  0.00 bits/sec    4   22.6 KBytes
[  4] 466.00-467.00 sec  0.00 Bytes  0.00 bits/sec    5   29.7 KBytes
[  4] 467.00-468.00 sec  60.8 KBytes   498 Kbits/sec   26    8.48 KBytes
[  4] 468.00-469.00 sec  0.00 Bytes  0.00 bits/sec    4   14.1 KBytes
[  4] 469.00-470.00 sec  0.00 Bytes  0.00 bits/sec    5   21.2 KBytes
[  4] 470.00-471.00 sec  0.00 Bytes  0.00 bits/sec    4   26.9 KBytes
[  4] 471.00-472.00 sec  0.00 Bytes  0.00 bits/sec    4   32.5 KBytes
[  4] 472.00-473.00 sec  59.4 KBytes   486 Kbits/sec   28    9.90 KBytes
[  4] 473.00-474.00 sec  0.00 Bytes  0.00 bits/sec    4   15.6 KBytes
[  4] 474.00-475.00 sec  0.00 Bytes  0.00 bits/sec    5   22.6 KBytes
[  4] 475.00-476.00 sec  0.00 Bytes  0.00 bits/sec    4   28.3 KBytes
[  4] 476.00-477.00 sec  59.4 KBytes   487 Kbits/sec   26    5.66 KBytes
[  4] 477.00-478.00 sec  0.00 Bytes  0.00 bits/sec    5   12.7 KBytes
[  4] 478.00-479.00 sec  0.00 Bytes  0.00 bits/sec    4   18.4 KBytes
[  4] 479.00-480.00 sec  0.00 Bytes  0.00 bits/sec    5   25.5 KBytes
[  4] 480.00-481.00 sec  0.00 Bytes  0.00 bits/sec    4   31.1 KBytes
[  4] 481.00-482.00 sec  59.4 KBytes   486 Kbits/sec   26    8.48 KBytes
[  4] 482.00-483.00 sec  0.00 Bytes  0.00 bits/sec    5   15.6 KBytes
[  4] 483.00-484.00 sec  0.00 Bytes  0.00 bits/sec    4   21.2 KBytes
[  4] 484.00-485.00 sec  0.00 Bytes  0.00 bits/sec    5   28.3 KBytes
[  4] 485.00-486.00 sec  62.2 KBytes   510 Kbits/sec   26    5.66 KBytes
[  4] 486.00-487.00 sec  0.00 Bytes  0.00 bits/sec    4   11.3 KBytes
[  4] 487.00-488.00 sec  0.00 Bytes  0.00 bits/sec    5   18.4 KBytes
[  4] 488.00-489.00 sec  0.00 Bytes  0.00 bits/sec    4   24.0 KBytes
[  4] 489.00-490.00 sec  0.00 Bytes  0.00 bits/sec    5   29.7 KBytes
[  4] 490.00-491.00 sec  59.4 KBytes   487 Kbits/sec   27    7.07 KBytes
[  4] 491.00-492.00 sec  0.00 Bytes  0.00 bits/sec    4   12.7 KBytes
[  4] 492.00-493.00 sec  0.00 Bytes  0.00 bits/sec    5   19.8 KBytes
[  4] 493.00-494.00 sec  0.00 Bytes  0.00 bits/sec    4   25.5 KBytes
[  4] 494.00-495.00 sec  0.00 Bytes  0.00 bits/sec    4   31.1 KBytes
[  4] 495.00-496.00 sec  59.4 KBytes   487 Kbits/sec   27    9.90 KBytes
[  4] 496.00-497.00 sec  0.00 Bytes  0.00 bits/sec    4   15.6 KBytes
[  4] 497.00-498.00 sec  0.00 Bytes  0.00 bits/sec    5   22.6 KBytes
[  4] 498.00-499.00 sec  0.00 Bytes  0.00 bits/sec    4   28.3 KBytes
[  4] 499.00-500.00 sec  0.00 Bytes  0.00 bits/sec    6   2.83 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth       Retr
[  4]   0.00-500.00 sec  7.10 MBytes   119 Kbits/sec  4367             sender
[  4]   0.00-500.00 sec  7.07 MBytes   119 Kbits/sec                  receiver

iperf Done.
root@n7:/tmp/pycore.39760/n7.conf# traceroute 10.0.13.20
traceroute to 10.0.13.20 (10.0.13.20), 30 hops max, 60 byte packets
 1  10.0.12.1 (10.0.12.1)  0.510 ms  0.437 ms  0.462 ms
 2  10.0.5.2 (10.0.5.2)  1.466 ms  1.511 ms  1.547 ms
 3  * * *
 4  10.0.13.20 (10.0.13.20)  20.558 ms  21.141 ms  20.906 ms
root@n7:/tmp/pycore.39760/n7.conf#
```

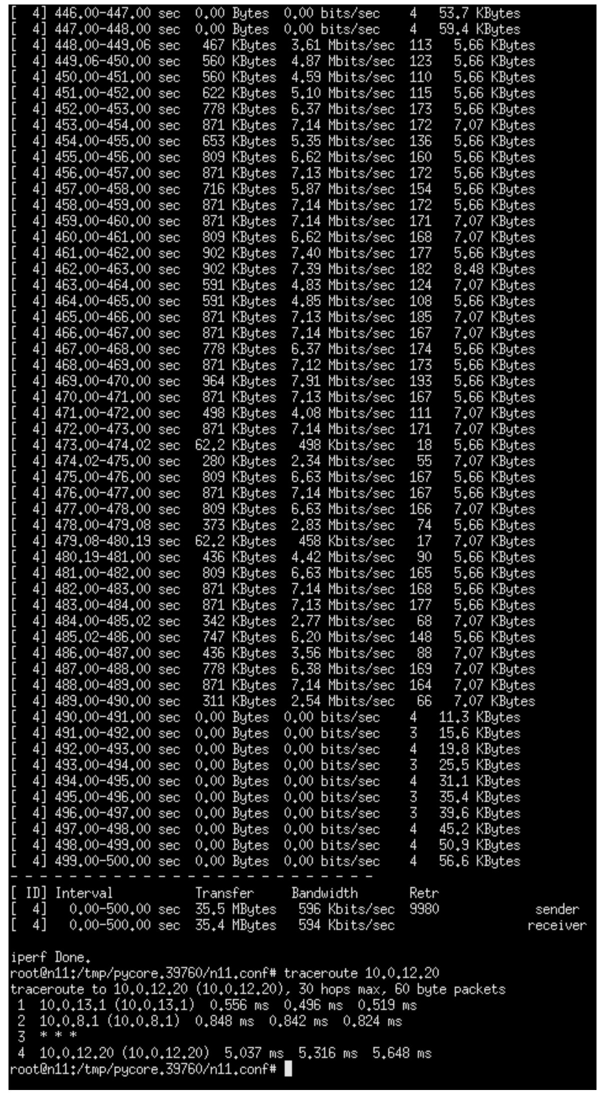
Figure 5: Results from step 3 (iperf3)

4.2 Question 5.ii



```
[  4] 446.00-447.00 sec   356 KBytes  2.92 Mbits/sec   74   2.83 KBytes
[  4] 447.00-448.00 sec   416 KBytes  3.40 Mbits/sec   77   2.83 KBytes
[  4] 448.00-449.00 sec   238 KBytes  1.94 Mbits/sec   52   4.24 KBytes
[  4] 449.00-450.00 sec   238 KBytes  1.95 Mbits/sec   44   4.24 KBytes
[  4] 450.00-451.00 sec   356 KBytes  2.92 Mbits/sec   78   2.83 KBytes
[  4] 451.00-452.00 sec   356 KBytes  2.92 Mbits/sec   69   2.83 KBytes
[  4] 452.00-453.00 sec   356 KBytes  2.92 Mbits/sec   72   2.83 KBytes
[  4] 453.00-454.00 sec   356 KBytes  2.92 Mbits/sec   73   2.83 KBytes
[  4] 454.00-455.00 sec   416 KBytes  3.41 Mbits/sec   80   4.24 KBytes
[  4] 455.00-456.00 sec   297 KBytes  2.43 Mbits/sec   70   2.83 KBytes
[  4] 456.00-457.00 sec   238 KBytes  1.95 Mbits/sec   46   2.83 KBytes
[  4] 457.00-458.00 sec   356 KBytes  2.92 Mbits/sec   71   2.83 KBytes
[  4] 458.00-459.00 sec  59.4 KBytes   487 Kbits/sec    9   4.24 KBytes
[  4] 459.00-460.00 sec   119 KBytes   973 Kbits/sec   22   4.24 KBytes
[  4] 460.00-461.00 sec   297 KBytes  2.43 Mbits/sec   69   2.83 KBytes
[  4] 461.00-462.00 sec   356 KBytes  2.92 Mbits/sec   70   2.83 KBytes
[  4] 462.00-463.00 sec   356 KBytes  2.92 Mbits/sec   69   4.24 KBytes
[  4] 463.00-464.00 sec   178 KBytes  1.46 Mbits/sec   34   2.83 KBytes
[  4] 464.00-465.00 sec  0.00 Bytes  0.00 bits/sec    7   2.83 KBytes
[  4] 465.00-466.00 sec   178 KBytes  1.46 Mbits/sec   37   4.24 KBytes
[  4] 466.00-467.00 sec   356 KBytes  2.92 Mbits/sec   68   4.24 KBytes
[  4] 467.00-468.00 sec   356 KBytes  2.92 Mbits/sec   71   2.83 KBytes
[  4] 468.00-469.00 sec   356 KBytes  2.92 Mbits/sec   73   4.24 KBytes
[  4] 469.00-470.00 sec   178 KBytes  1.46 Mbits/sec   31   2.83 KBytes
[  4] 470.00-471.00 sec   297 KBytes  2.44 Mbits/sec   61   2.83 KBytes
[  4] 471.00-472.00 sec   178 KBytes  1.46 Mbits/sec   38   4.24 KBytes
[  4] 472.00-473.00 sec   356 KBytes  2.92 Mbits/sec   70   2.83 KBytes
[  4] 473.00-474.00 sec   297 KBytes  2.43 Mbits/sec   68   4.24 KBytes
[  4] 474.00-475.00 sec   178 KBytes  1.46 Mbits/sec   34   2.83 KBytes
[  4] 475.00-476.00 sec   356 KBytes  2.92 Mbits/sec   75   2.83 KBytes
[  4] 476.00-477.00 sec   416 KBytes  3.41 Mbits/sec   81   4.24 KBytes
[  4] 477.00-478.00 sec   178 KBytes  1.46 Mbits/sec   36   4.24 KBytes
[  4] 478.00-479.00 sec   356 KBytes  2.92 Mbits/sec   76   2.83 KBytes
[  4] 479.00-480.00 sec   178 KBytes  1.46 Mbits/sec   36   2.83 KBytes
[  4] 480.00-481.00 sec   119 KBytes   975 Kbits/sec   10   4.24 KBytes
[  4] 481.00-482.00 sec  0.00 Bytes  0.00 bits/sec    8   8.48 KBytes
[  4] 482.00-483.00 sec  59.4 KBytes   487 Kbits/sec   10   8.48 KBytes
[  4] 483.00-484.00 sec  0.00 Bytes  0.00 bits/sec   12   8.48 KBytes
[  4] 484.00-485.00 sec  0.00 Bytes  0.00 bits/sec    4   14.1 KBytes
[  4] 485.00-486.00 sec  0.00 Bytes  0.00 bits/sec    5   21.2 KBytes
[  4] 486.00-487.00 sec  0.00 Bytes  0.00 bits/sec    4   26.9 KBytes
[  4] 487.00-488.00 sec  59.4 KBytes   486 Kbits/sec   23   11.3 KBytes
[  4] 488.00-489.00 sec  0.00 Bytes  0.00 bits/sec    4   17.0 KBytes
[  4] 489.00-490.00 sec  0.00 Bytes  0.00 bits/sec    5   24.0 KBytes
[  4] 490.00-491.00 sec  59.4 KBytes   486 Kbits/sec   23   5.66 KBytes
[  4] 491.00-492.00 sec  0.00 Bytes  0.00 bits/sec    5   12.7 KBytes
[  4] 492.00-493.00 sec  59.4 KBytes   487 Kbits/sec   13   8.48 KBytes
[  4] 493.00-494.00 sec  0.00 Bytes  0.00 bits/sec    5   15.6 KBytes
[  4] 494.00-495.00 sec  0.00 Bytes  0.00 bits/sec    4   21.2 KBytes
[  4] 495.00-496.00 sec  0.00 Bytes  0.00 bits/sec    5   28.3 KBytes
[  4] 496.00-497.00 sec  59.4 KBytes   487 Kbits/sec   23   9.90 KBytes
[  4] 497.00-498.00 sec  0.00 Bytes  0.00 bits/sec    5   17.0 KBytes
[  4] 498.00-499.00 sec  0.00 Bytes  0.00 bits/sec    4   22.6 KBytes
[  4] 499.00-500.00 sec  0.00 Bytes  0.00 bits/sec   24   5.66 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth       Retr
[  4]   0.00-500.00 sec  25.6 MBytes   429 Kbits/sec  7766             sender
[  4]   0.00-500.00 sec  25.5 MBytes   428 Kbits/sec                  receiver

iperf Done.
root@n7:/tmp/pycore.39760/n7.conf# traceroute 10.0.13.20
traceroute to 10.0.13.20 (10.0.13.20), 30 hops max, 60 byte packets
 1  10.0.12.1 (10.0.12.1)  0.483 ms  0.409 ms  0.395 ms
 2  10.0.4.1 (10.0.4.1)  0.905 ms  0.905 ms  0.990 ms
 3  * * *
 4  10.0.13.20 (10.0.13.20)  4.279 ms * *
root@n7:/tmp/pycore.39760/n7.conf#
```

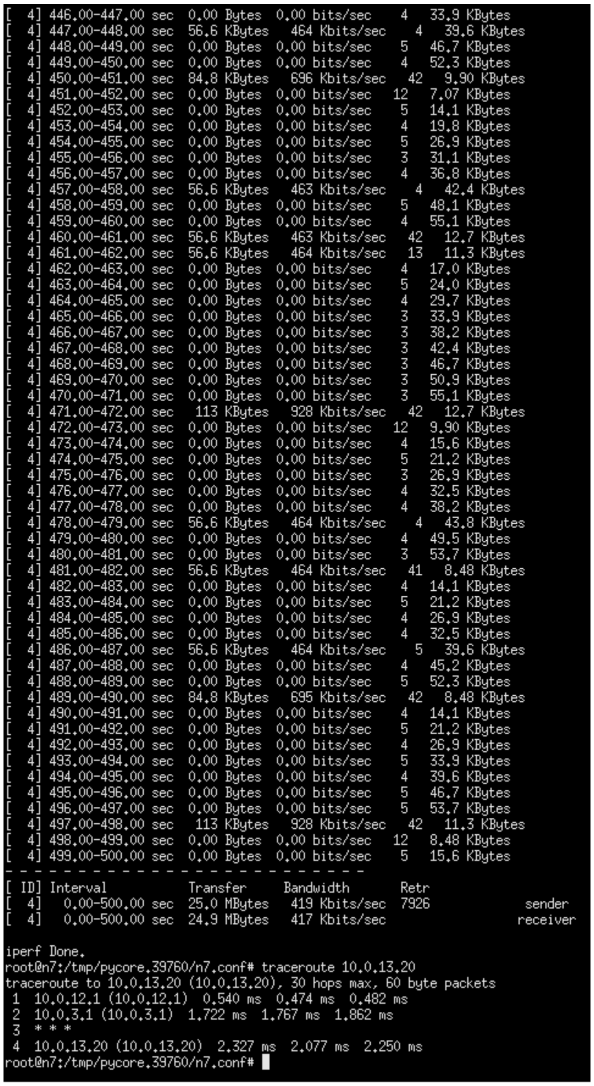
(a) Node 7 to Node 11 (cost=10)



```
[  4] 446.00-447.00 sec  0.00 Bytes  0.00 bits/sec    4   53.7 KBytes
[  4] 447.00-448.00 sec  0.00 Bytes  0.00 bits/sec    4   59.4 KBytes
[  4] 448.00-449.06 sec   467 KBytes  3.61 Mbits/sec  113   5.66 KBytes
[  4] 449.06-450.00 sec   560 KBytes  4.87 Mbits/sec  123   5.66 KBytes
[  4] 450.00-451.00 sec   560 KBytes  4.59 Mbits/sec  110   5.66 KBytes
[  4] 451.00-452.00 sec   622 KBytes  5.10 Mbits/sec  115   5.66 KBytes
[  4] 452.00-453.00 sec   778 KBytes  6.37 Mbits/sec  173   5.66 KBytes
[  4] 453.00-454.00 sec   871 KBytes  7.14 Mbits/sec  172   7.07 KBytes
[  4] 454.00-455.00 sec   653 KBytes  5.35 Mbits/sec  136   5.66 KBytes
[  4] 455.00-456.00 sec   809 KBytes  6.62 Mbits/sec  160   5.66 KBytes
[  4] 456.00-457.00 sec   871 KBytes  7.13 Mbits/sec  172   5.66 KBytes
[  4] 457.00-458.00 sec   716 KBytes  5.87 Mbits/sec  154   5.66 KBytes
[  4] 458.00-459.00 sec   871 KBytes  7.14 Mbits/sec  172   5.66 KBytes
[  4] 459.00-460.00 sec   871 KBytes  7.14 Mbits/sec  171   7.07 KBytes
[  4] 460.00-461.00 sec   809 KBytes  6.62 Mbits/sec  168   7.07 KBytes
[  4] 461.00-462.00 sec   902 KBytes  7.40 Mbits/sec  177   5.66 KBytes
[  4] 462.00-463.00 sec   902 KBytes  7.39 Mbits/sec  182   8.48 KBytes
[  4] 463.00-464.00 sec   591 KBytes  4.83 Mbits/sec  124   7.07 KBytes
[  4] 464.00-465.00 sec   591 KBytes  4.85 Mbits/sec  108   5.66 KBytes
[  4] 465.00-466.00 sec   871 KBytes  7.13 Mbits/sec  185   7.07 KBytes
[  4] 466.00-467.00 sec   871 KBytes  7.14 Mbits/sec  167   7.07 KBytes
[  4] 467.00-468.00 sec   778 KBytes  6.37 Mbits/sec  174   5.66 KBytes
[  4] 468.00-469.00 sec   871 KBytes  7.12 Mbits/sec  173   5.66 KBytes
[  4] 469.00-470.00 sec   964 KBytes  7.91 Mbits/sec  193   5.66 KBytes
[  4] 470.00-471.00 sec   871 KBytes  7.13 Mbits/sec  167   5.66 KBytes
[  4] 471.00-472.00 sec   498 KBytes  4.08 Mbits/sec  111   7.07 KBytes
[  4] 472.00-473.00 sec   871 KBytes  7.14 Mbits/sec  171   7.07 KBytes
[  4] 473.00-474.02 sec  62.2 KBytes   498 Kbits/sec   18   5.66 KBytes
[  4] 474.02-475.00 sec   280 KBytes  2.34 Mbits/sec   55   7.07 KBytes
[  4] 475.00-476.00 sec   809 KBytes  6.63 Mbits/sec  167   5.66 KBytes
[  4] 476.00-477.00 sec   871 KBytes  7.14 Mbits/sec  167   5.66 KBytes
[  4] 477.00-478.00 sec   809 KBytes  6.63 Mbits/sec  166   7.07 KBytes
[  4] 478.00-479.08 sec   373 KBytes  2.83 Mbits/sec   74   5.66 KBytes
[  4] 479.08-480.19 sec  62.2 KBytes   458 Kbits/sec   17   7.07 KBytes
[  4] 480.19-481.00 sec   436 KBytes  4.42 Mbits/sec   90   5.66 KBytes
[  4] 481.00-482.00 sec   809 KBytes  6.63 Mbits/sec  165   5.66 KBytes
[  4] 482.00-483.00 sec   871 KBytes  7.14 Mbits/sec  168   5.66 KBytes
[  4] 483.00-484.00 sec   871 KBytes  7.13 Mbits/sec  177   5.66 KBytes
[  4] 484.00-485.02 sec   342 KBytes  2.77 Mbits/sec   68   7.07 KBytes
[  4] 485.02-486.00 sec   747 KBytes  6.20 Mbits/sec  148   5.66 KBytes
[  4] 486.00-487.00 sec   436 KBytes  3.56 Mbits/sec   88   7.07 KBytes
[  4] 487.00-488.00 sec   778 KBytes  6.38 Mbits/sec  169   7.07 KBytes
[  4] 488.00-489.00 sec   871 KBytes  7.14 Mbits/sec  164   7.07 KBytes
[  4] 489.00-490.00 sec   311 KBytes  2.54 Mbits/sec   66   7.07 KBytes
[  4] 490.00-491.00 sec  0.00 Bytes  0.00 bits/sec    4   11.3 KBytes
[  4] 491.00-492.00 sec  0.00 Bytes  0.00 bits/sec    3   15.6 KBytes
[  4] 492.00-493.00 sec  0.00 Bytes  0.00 bits/sec    4   19.8 KBytes
[  4] 493.00-494.00 sec  0.00 Bytes  0.00 bits/sec    3   25.5 KBytes
[  4] 494.00-495.00 sec  0.00 Bytes  0.00 bits/sec    4   31.1 KBytes
[  4] 495.00-496.00 sec  0.00 Bytes  0.00 bits/sec    3   35.4 KBytes
[  4] 496.00-497.00 sec  0.00 Bytes  0.00 bits/sec    3   39.6 KBytes
[  4] 497.00-498.00 sec  0.00 Bytes  0.00 bits/sec    4   45.2 KBytes
[  4] 498.00-499.00 sec  0.00 Bytes  0.00 bits/sec    4   50.9 KBytes
[  4] 499.00-500.00 sec  0.00 Bytes  0.00 bits/sec    4   56.6 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth       Retr
[  4]   0.00-500.00 sec  35.5 MBytes   596 Kbits/sec  9980             sender
[  4]   0.00-500.00 sec  35.4 MBytes   594 Kbits/sec                  receiver

iperf Done.
root@n11:/tmp/pycore.39760/n11.conf# traceroute 10.0.12.20
traceroute to 10.0.12.20 (10.0.12.20), 30 hops max, 60 byte packets
 1  10.0.13.1 (10.0.13.1)  0.556 ms  0.496 ms  0.519 ms
 2  10.0.8.1 (10.0.8.1)  0.848 ms  0.842 ms  0.824 ms
 3  * * *
 4  10.0.12.20 (10.0.12.20)  5.037 ms  5.316 ms  5.648 ms
root@n11:/tmp/pycore.39760/n11.conf#
```

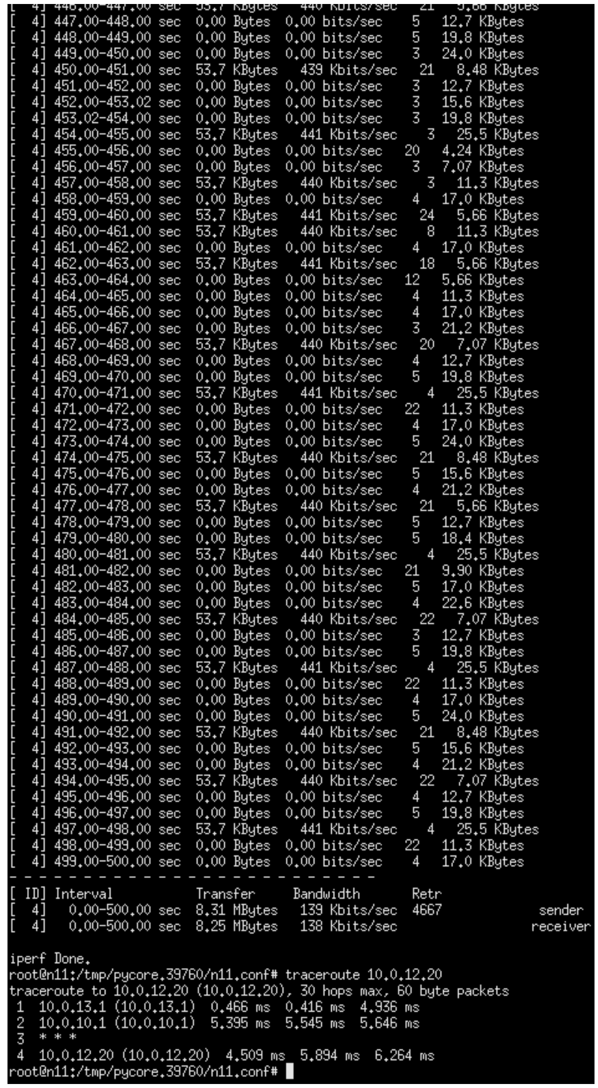
(b) Node 11 to Node 7 (cost=10)

Figure 6: Network performance with cost = 10



```
[  4] 446.00-447.00 sec  0.00 Bytes  0.00 bits/sec    4   33.9 KBytes
[  4] 447.00-448.00 sec  56.6 KBytes   464 Kbits/sec    4   39.6 KBytes
[  4] 448.00-449.00 sec  0.00 Bytes  0.00 bits/sec    5   46.7 KBytes
[  4] 449.00-450.00 sec  0.00 Bytes  0.00 bits/sec    4   52.3 KBytes
[  4] 450.00-451.00 sec  84.8 KBytes   696 Kbits/sec   42   9.90 KBytes
[  4] 451.00-452.00 sec  0.00 Bytes  0.00 bits/sec   12   7.07 KBytes
[  4] 452.00-453.00 sec  0.00 Bytes  0.00 bits/sec    5   14.1 KBytes
[  4] 453.00-454.00 sec  0.00 Bytes  0.00 bits/sec    4   19.8 KBytes
[  4] 454.00-455.00 sec  0.00 Bytes  0.00 bits/sec    5   26.9 KBytes
[  4] 455.00-456.00 sec  0.00 Bytes  0.00 bits/sec    3   31.1 KBytes
[  4] 456.00-457.00 sec  0.00 Bytes  0.00 bits/sec    4   36.8 KBytes
[  4] 457.00-458.00 sec  56.6 KBytes   463 Kbits/sec    4   42.4 KBytes
[  4] 458.00-459.00 sec  0.00 Bytes  0.00 bits/sec    5   48.1 KBytes
[  4] 459.00-460.00 sec  0.00 Bytes  0.00 bits/sec    4   55.1 KBytes
[  4] 460.00-461.00 sec  56.6 KBytes   463 Kbits/sec   42   12.7 KBytes
[  4] 461.00-462.00 sec  56.6 KBytes   464 Kbits/sec   13   11.3 KBytes
[  4] 462.00-463.00 sec  0.00 Bytes  0.00 bits/sec    4   17.0 KBytes
[  4] 463.00-464.00 sec  0.00 Bytes  0.00 bits/sec    5   24.0 KBytes
[  4] 464.00-465.00 sec  0.00 Bytes  0.00 bits/sec    4   29.7 KBytes
[  4] 465.00-466.00 sec  0.00 Bytes  0.00 bits/sec    3   33.9 KBytes
[  4] 466.00-467.00 sec  0.00 Bytes  0.00 bits/sec    3   38.2 KBytes
[  4] 467.00-468.00 sec  0.00 Bytes  0.00 bits/sec    3   42.4 KBytes
[  4] 468.00-469.00 sec  0.00 Bytes  0.00 bits/sec    3   46.7 KBytes
[  4] 469.00-470.00 sec  0.00 Bytes  0.00 bits/sec    3   50.9 KBytes
[  4] 470.00-471.00 sec  0.00 Bytes  0.00 bits/sec    3   55.1 KBytes
[  4] 471.00-472.00 sec   113 KBytes   928 Kbits/sec   42   12.7 KBytes
[  4] 472.00-473.00 sec  0.00 Bytes  0.00 bits/sec   12   9.90 KBytes
[  4] 473.00-474.00 sec  0.00 Bytes  0.00 bits/sec    4   15.6 KBytes
[  4] 474.00-475.00 sec  0.00 Bytes  0.00 bits/sec    5   21.2 KBytes
[  4] 475.00-476.00 sec  0.00 Bytes  0.00 bits/sec    3   26.9 KBytes
[  4] 476.00-477.00 sec  0.00 Bytes  0.00 bits/sec    4   32.5 KBytes
[  4] 477.00-478.00 sec  0.00 Bytes  0.00 bits/sec    4   38.2 KBytes
[  4] 478.00-479.00 sec  56.6 KBytes   464 Kbits/sec    4   43.8 KBytes
[  4] 479.00-480.00 sec  0.00 Bytes  0.00 bits/sec    4   49.5 KBytes
[  4] 480.00-481.00 sec  0.00 Bytes  0.00 bits/sec    3   53.7 KBytes
[  4] 481.00-482.00 sec  56.6 KBytes   464 Kbits/sec   41   8.48 KBytes
[  4] 482.00-483.00 sec  0.00 Bytes  0.00 bits/sec    4   14.1 KBytes
[  4] 483.00-484.00 sec  0.00 Bytes  0.00 bits/sec    5   21.2 KBytes
[  4] 484.00-485.00 sec  0.00 Bytes  0.00 bits/sec    4   26.9 KBytes
[  4] 485.00-486.00 sec  0.00 Bytes  0.00 bits/sec    4   32.5 KBytes
[  4] 486.00-487.00 sec  56.6 KBytes   464 Kbits/sec    5   39.6 KBytes
[  4] 487.00-488.00 sec  0.00 Bytes  0.00 bits/sec    4   45.2 KBytes
[  4] 488.00-489.00 sec  0.00 Bytes  0.00 bits/sec    5   52.3 KBytes
[  4] 489.00-490.00 sec  84.8 KBytes   695 Kbits/sec   42   8.48 KBytes
[  4] 490.00-491.00 sec  0.00 Bytes  0.00 bits/sec    4   14.1 KBytes
[  4] 491.00-492.00 sec  0.00 Bytes  0.00 bits/sec    5   21.2 KBytes
[  4] 492.00-493.00 sec  0.00 Bytes  0.00 bits/sec    4   26.9 KBytes
[  4] 493.00-494.00 sec  0.00 Bytes  0.00 bits/sec    5   33.9 KBytes
[  4] 494.00-495.00 sec  0.00 Bytes  0.00 bits/sec    4   39.6 KBytes
[  4] 495.00-496.00 sec  0.00 Bytes  0.00 bits/sec    5   46.7 KBytes
[  4] 496.00-497.00 sec  0.00 Bytes  0.00 bits/sec    5   53.7 KBytes
[  4] 497.00-498.00 sec   113 KBytes   928 Kbits/sec   42   11.3 KBytes
[  4] 498.00-499.00 sec  0.00 Bytes  0.00 bits/sec   12   8.48 KBytes
[  4] 499.00-500.00 sec  0.00 Bytes  0.00 bits/sec    5   15.6 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth       Retr
[  4]   0.00-500.00 sec  25.0 MBytes   419 Kbits/sec  7926             sender
[  4]   0.00-500.00 sec  24.9 MBytes   417 Kbits/sec                  receiver

iperf Done.
root@n7:/tmp/pycore.39760/n7.conf# traceroute 10.0.13.20
traceroute to 10.0.13.20 (10.0.13.20), 30 hops max, 60 byte packets
 1  10.0.12.1 (10.0.12.1)  0.540 ms  0.474 ms  0.482 ms
 2  10.0.3.1 (10.0.3.1)  1.722 ms  1.767 ms  1.862 ms
 3  * * *
 4  10.0.13.20 (10.0.13.20)  2.327 ms  2.077 ms  2.250 ms
root@n7:/tmp/pycore.39760/n7.conf#
```

(a) Node 7 to Node 11 (cost=40)



```
[  4] 447.00-448.00 sec  0.00 Bytes  0.00 bits/sec    5   12.7 KBytes
[  4] 448.00-449.00 sec  0.00 Bytes  0.00 bits/sec    5   19.8 KBytes
[  4] 449.00-450.00 sec  0.00 Bytes  0.00 bits/sec    3   24.0 KBytes
[  4] 450.00-451.00 sec  53.7 KBytes   439 Kbits/sec   21   8.48 KBytes
[  4] 451.00-452.00 sec  0.00 Bytes  0.00 bits/sec    3   12.7 KBytes
[  4] 452.00-453.02 sec  0.00 Bytes  0.00 bits/sec    3   15.6 KBytes
[  4] 453.02-454.00 sec  0.00 Bytes  0.00 bits/sec    3   19.8 KBytes
[  4] 454.00-455.00 sec  53.7 KBytes   441 Kbits/sec    3   25.5 KBytes
[  4] 455.00-456.00 sec  0.00 Bytes  0.00 bits/sec   20   4.24 KBytes
[  4] 456.00-457.00 sec  0.00 Bytes  0.00 bits/sec    3   7.07 KBytes
[  4] 457.00-458.00 sec  53.7 KBytes   440 Kbits/sec    3   11.3 KBytes
[  4] 458.00-459.00 sec  0.00 Bytes  0.00 bits/sec    4   17.0 KBytes
[  4] 459.00-460.00 sec  53.7 KBytes   441 Kbits/sec   24   5.66 KBytes
[  4] 460.00-461.00 sec  53.7 KBytes   440 Kbits/sec    8   11.3 KBytes
[  4] 461.00-462.00 sec  0.00 Bytes  0.00 bits/sec    4   17.0 KBytes
[  4] 462.00-463.00 sec  53.7 KBytes   441 Kbits/sec   18   5.66 KBytes
[  4] 463.00-464.00 sec  0.00 Bytes  0.00 bits/sec   12   5.66 KBytes
[  4] 464.00-465.00 sec  0.00 Bytes  0.00 bits/sec    4   11.3 KBytes
[  4] 465.00-466.00 sec  0.00 Bytes  0.00 bits/sec    4   17.0 KBytes
[  4] 466.00-467.00 sec  0.00 Bytes  0.00 bits/sec    3   21.2 KBytes
[  4] 467.00-468.00 sec  53.7 KBytes   440 Kbits/sec   20   7.07 KBytes
[  4] 468.00-469.00 sec  0.00 Bytes  0.00 bits/sec    4   12.7 KBytes
[  4] 469.00-470.00 sec  0.00 Bytes  0.00 bits/sec    5   19.8 KBytes
[  4] 470.00-471.00 sec  53.7 KBytes   441 Kbits/sec    4   25.5 KBytes
[  4] 471.00-472.00 sec  0.00 Bytes  0.00 bits/sec   22   11.3 KBytes
[  4] 472.00-473.00 sec  0.00 Bytes  0.00 bits/sec    4   17.0 KBytes
[  4] 473.00-474.00 sec  0.00 Bytes  0.00 bits/sec    5   24.0 KBytes
[  4] 474.00-475.00 sec  53.7 KBytes   440 Kbits/sec   21   8.48 KBytes
[  4] 475.00-476.00 sec  0.00 Bytes  0.00 bits/sec    5   15.6 KBytes
[  4] 476.00-477.00 sec  0.00 Bytes  0.00 bits/sec    4   21.2 KBytes
[  4] 477.00-478.00 sec  53.7 KBytes   440 Kbits/sec   21   5.66 KBytes
[  4] 478.00-479.00 sec  0.00 Bytes  0.00 bits/sec    5   12.7 KBytes
[  4] 479.00-480.00 sec  0.00 Bytes  0.00 bits/sec    5   18.4 KBytes
[  4] 480.00-481.00 sec  53.7 KBytes   440 Kbits/sec    4   25.5 KBytes
[  4] 481.00-482.00 sec  0.00 Bytes  0.00 bits/sec   21   9.90 KBytes
[  4] 482.00-483.00 sec  0.00 Bytes  0.00 bits/sec    5   17.0 KBytes
[  4] 483.00-484.00 sec  0.00 Bytes  0.00 bits/sec    4   22.6 KBytes
[  4] 484.00-485.00 sec  53.7 KBytes   440 Kbits/sec   22   7.07 KBytes
[  4] 485.00-486.00 sec  0.00 Bytes  0.00 bits/sec    3   12.7 KBytes
[  4] 486.00-487.00 sec  0.00 Bytes  0.00 bits/sec    5   19.8 KBytes
[  4] 487.00-488.00 sec  53.7 KBytes   441 Kbits/sec    4   25.5 KBytes
[  4] 488.00-489.00 sec  0.00 Bytes  0.00 bits/sec   22   11.3 KBytes
[  4] 489.00-490.00 sec  0.00 Bytes  0.00 bits/sec    4   17.0 KBytes
[  4] 490.00-491.00 sec  0.00 Bytes  0.00 bits/sec    5   24.0 KBytes
[  4] 491.00-492.00 sec  53.7 KBytes   440 Kbits/sec   21   8.48 KBytes
[  4] 492.00-493.00 sec  0.00 Bytes  0.00 bits/sec    5   15.6 KBytes
[  4] 493.00-494.00 sec  0.00 Bytes  0.00 bits/sec    4   21.2 KBytes
[  4] 494.00-495.00 sec  53.7 KBytes   440 Kbits/sec   22   7.07 KBytes
[  4] 495.00-496.00 sec  0.00 Bytes  0.00 bits/sec    4   12.7 KBytes
[  4] 496.00-497.00 sec  0.00 Bytes  0.00 bits/sec    5   19.8 KBytes
[  4] 497.00-498.00 sec  53.7 KBytes   441 Kbits/sec    4   25.5 KBytes
[  4] 498.00-499.00 sec  0.00 Bytes  0.00 bits/sec   22   11.3 KBytes
[  4] 499.00-500.00 sec  0.00 Bytes  0.00 bits/sec    4   17.0 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth       Retr
[  4]   0.00-500.00 sec  8.31 MBytes   139 Kbits/sec  4667             sender
[  4]   0.00-500.00 sec  8.25 MBytes   138 Kbits/sec                  receiver

iperf Done.
root@n11:/tmp/pycore.39760/n11.conf# traceroute 10.0.12.20
traceroute to 10.0.12.20 (10.0.12.20), 30 hops max, 60 byte packets
 1  10.0.13.1 (10.0.13.1)  0.466 ms  0.416 ms  4.936 ms
 2  10.0.10.1 (10.0.10.1)  5.395 ms  5.545 ms  5.646 ms
 3  * * *
 4  10.0.12.20 (10.0.12.20)  4.509 ms  5.894 ms  6.264 ms
root@n11:/tmp/pycore.39760/n11.conf#
```

(b) Node 11 to Node 7 (cost=40)

Figure 7: Network performance with cost = 40

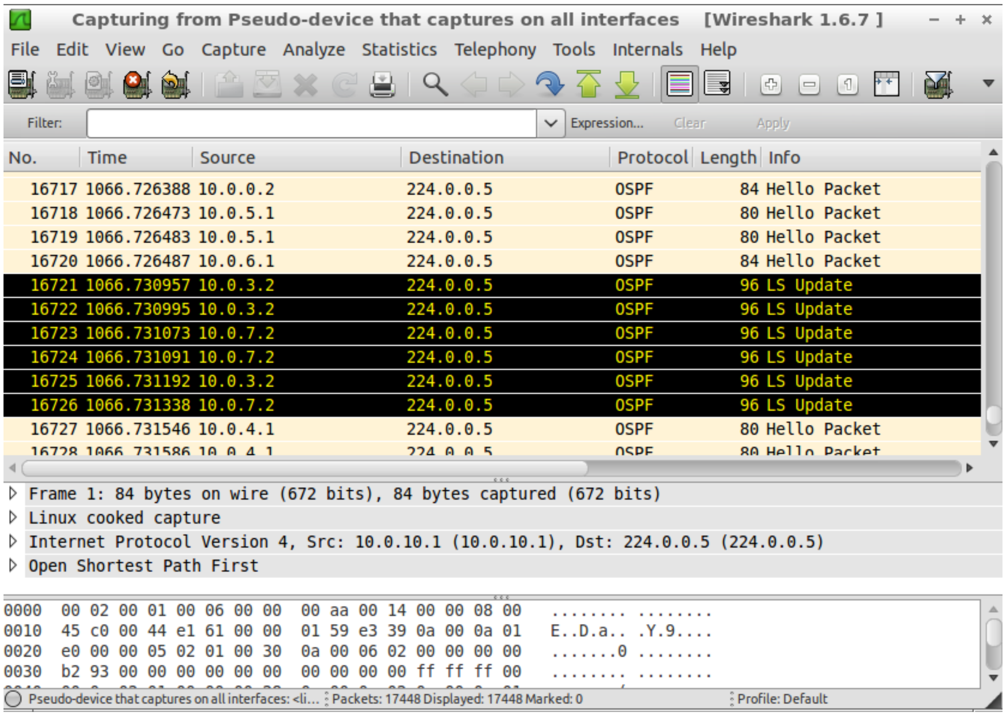
5 OSPF Database Updates

5.1 Question 6.i

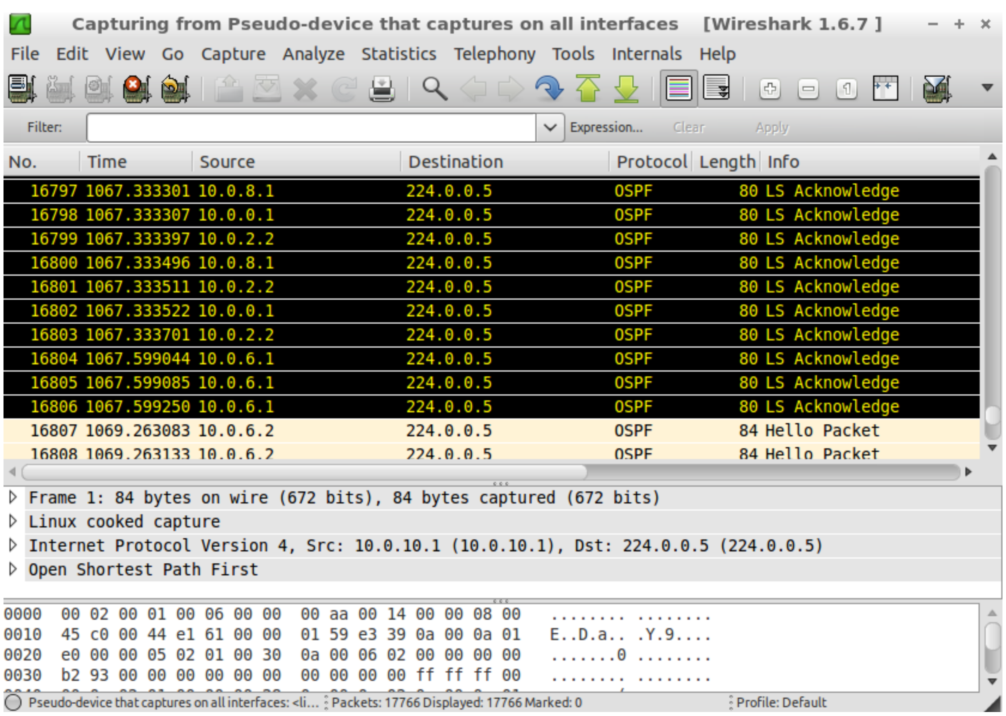
```
Hello, this is Quagga (version 0.99.20.1).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

n2# config terminal
n2(config)# show ip ospf database
% Unknown command.
n2(config)# do show ip ospf database

       OSPF Router with ID (10.0.1.2)

                Router Link States (Area 0.0.0.0)

Link ID         ADV Router      Age  Seq#       CkSum  Link count
10.0.0.1        10.0.0.1        1001 0x8000000f 0xe8c2 3
10.0.0.2        10.0.0.2        1081 0x80000010 0x8e0e 3
10.0.1.2        10.0.1.2        1050 0x8000000f 0x830f 3
10.0.3.1        10.0.3.1        1554 0x80000011 0x3931 3
10.0.3.2        10.0.3.2        1193 0x80000010 0xa1e4 3
10.0.5.1        10.0.5.1        1054 0x8000000e 0x880b 3
10.0.6.2        10.0.6.2        1160 0x8000000e 0xc2b0 3
10.0.8.2        10.0.8.2        1051 0x80000010 0x1a19 4

                Net Link States (Area 0.0.0.0)

Link ID         ADV Router      Age  Seq#       CkSum
10.0.0.2        10.0.0.2        1011 0x80000006 0x55d0
10.0.1.2        10.0.1.2         940 0x80000006 0x5ac7
10.0.2.1        10.0.1.2          70 0x80000007 0x45dc
10.0.3.2        10.0.3.2         813 0x80000006 0x61b8
10.0.4.2        10.0.5.1        1214 0x80000006 0x5abc

...skipping one line
10.0.6.2        10.0.6.2        1780 0x80000006 0x41d1
10.0.7.1        10.0.6.2         950 0x80000006 0x4fc0
10.0.8.2        10.0.8.2        1631 0x80000006 0x48c2
10.0.9.1        10.0.8.2        1011 0x80000006 0x23e8
10.0.10.2       10.0.8.2        1011 0x80000006 0x57ad


n2(config)# show ip ospf route
% Unknown command.
n2(config)# do show ip ospf route
============ OSPF network routing table ============
N    10.0.0.0/24           [20] area: 0.0.0.0
                           via 10.0.1.1, eth0
                           via 10.0.2.2, eth1
N    10.0.1.0/24           [10] area: 0.0.0.0
                           directly attached to eth0
N    10.0.2.0/24           [10] area: 0.0.0.0
                           directly attached to eth1
N    10.0.3.0/24           [30] area: 0.0.0.0
                           via 10.0.9.1, eth2
N    10.0.4.0/24           [50] area: 0.0.0.0
                           via 10.0.1.1, eth0
                           via 10.0.9.1, eth2
N    10.0.5.0/24           [40] area: 0.0.0.0
                           via 10.0.1.1, eth0
                           via 10.0.9.1, eth2
N    10.0.6.0/24           [20] area: 0.0.0.0
                           via 10.0.1.1, eth0
N    10.0.7.0/24           [30] area: 0.0.0.0
                           via 10.0.1.1, eth0
                           via 10.0.9.1, eth2
N    10.0.8.0/24           [20] area: 0.0.0.0
                           via 10.0.9.1, eth2
--More--
```

Figure 8: OSPF Database and Routing Table Results from Router n2

5.2 Question 6.ii



(a) First OSPF Update Message



(b) Last ACK Message

Figure 9: OSPF Link State Exchange Timing Analysis

5.3 Question 6.iii

```
Hello, this is Quagga (version 0.99.20.1).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

n2# config terminal
VTY configuration is locked by other VTY
n2# config terminal
n2(config)# do show ip ospf database

       OSPF Router with ID (10.0.1.2)

                Router Link States (Area 0.0.0.0)

Link ID         ADV Router      Age  Seq#       CkSum  Link count
10.0.0.1        10.0.0.1          21 0x80000010 0xe6c3 3
10.0.0.2        10.0.0.2         161 0x80000011 0x8c0f 3
10.0.1.2        10.0.1.2         140 0x80000010 0x8110 3
10.0.3.1        10.0.3.1          81 0x80000016 0xaa47 1
10.0.3.2        10.0.3.2          63 0x80000012 0xe6ad 3
10.0.5.1        10.0.5.1          44 0x80000010 0xf3ae 3
10.0.6.2        10.0.6.2         179 0x8000000f 0xc0b1 3
10.0.8.2        10.0.8.2          40 0x80000012 0x1e28 4

                Net Link States (Area 0.0.0.0)

Link ID         ADV Router      Age  Seq#       CkSum
10.0.0.2        10.0.0.2         141 0x80000007 0x53d1
10.0.1.2        10.0.1.2        1820 0x80000006 0x5ac7
10.0.2.1        10.0.1.2         950 0x80000007 0x45dc
10.0.5.1        10.0.5.1        1743 0x80000006 0x5fb6
10.0.6.2        10.0.6.2         840 0x80000007 0x3fd2
10.0.7.1        10.0.6.2           9 0x80000007 0x4dc1
10.0.9.1        10.0.8.2          61 0x80000007 0x21e9
10.0.10.2       10.0.8.2         111 0x80000007 0x55ae


n2(config)# do show ospf route
% Unknown command.
n2(config)# do show ip ospf route
============ OSPF network routing table ============
N    10.0.0.0/24           [20] area: 0.0.0.0
                           via 10.0.1.1, eth0
                           via 10.0.2.2, eth1
N    10.0.1.0/24           [10] area: 0.0.0.0
                           directly attached to eth0
N    10.0.2.0/24           [10] area: 0.0.0.0
                           directly attached to eth1
N    10.0.3.0/24           [40] area: 0.0.0.0
                           via 10.0.1.1, eth0
                           via 10.0.9.1, eth2
N    10.0.4.0/24           [50] area: 0.0.0.0
                           via 10.0.1.1, eth0
                           via 10.0.9.1, eth2
N    10.0.5.0/24           [40] area: 0.0.0.0
                           via 10.0.1.1, eth0
                           via 10.0.9.1, eth2
N    10.0.6.0/24           [20] area: 0.0.0.0
                           via 10.0.1.1, eth0
N    10.0.7.0/24           [30] area: 0.0.0.0
                           via 10.0.1.1, eth0
                           via 10.0.9.1, eth2
N    10.0.8.0/24           [20] area: 0.0.0.0
--More--
```

Figure 10: New Routing Table